MAHARASHTRA STATE BOARD OF TECHNICAL, EDUCATION

MUMBAI

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Synopsis On

“Develop a Gesture Recognition software for speech or hearing impairment peoples.”

Submitted By

1. Swapnil Sunil Dalvi (21121300).
2. Rushikesh Ranjeet Powar (2112130074).
3. Sanskruti Suresh Patil (2112130086).
4. Vishwaja Mallikarjun Pujari (2112130048).

Guided By: Mr. S.A. Mujawar.

Department of Computer Engineering

Dr. Bapuji Salunkhe Institute of Engineering and Technology,

Kolhapur

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1.0 Topic of the project:

Develop a Gesture Recognition for peoples having speech or hearing impairments.

2.0 Why is the particular topic chosen?

The selection of the topic "Developing a Gesture Recognition System for individuals with speech or hearing impairments" is motivated by a profound commitment to inclusivity, accessibility, and technological innovation. People with speech or hearing impairments often face formidable communication barriers that hinder their ability to interact with the world around them. By embarking on this project, we aim to bridge these gaps and provide a vital lifeline for those in need. This endeavor recognizes the pressing need for alternative communication solutions that empower individuals with disabilities, ensuring they have equal access to opportunities and social inclusion. Furthermore, it delves into the dynamic realm of gesture recognition, a technologically advanced field that combines computer vision, machine learning, and human-computer interaction. Through research, development, and innovation, this project aspires to improve upon existing solutions, offering a more accessible and efficient means of communication for this underserved community. Ultimately, it seeks to enrich lives, bolster independence, and catalyze personal growth while advancing the frontiers of assistive technology.

3.0 Objective and Scope of the Project:

1. Objective of the Project:

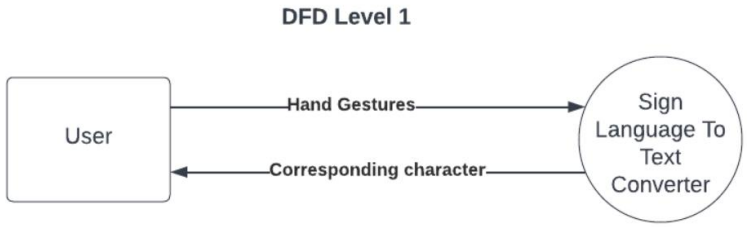
The primary objectives of the project, "Developing a Gesture Recognition System for individuals with speech or hearing impairments," are rooted in the core values of inclusivity, accessibility, and technological advancement. At its heart, this project seeks to break down communication barriers faced by individuals with speech or hearing impairments. It strives to empower them with an alternative means of expression, providing them with the tools they need to communicate effectively with the world around them. By developing a precise and intuitive gesture recognition system, we aim to enhance their communication experience, ensuring that it is both efficient and accurate. This system's real-time capabilities will enable fluid and natural interactions, promoting meaningful conversations that may have otherwise been challenging. The project's extensive vocabulary and customizable features are designed to adapt to individual preferences and cater to a wide range of communication needs.

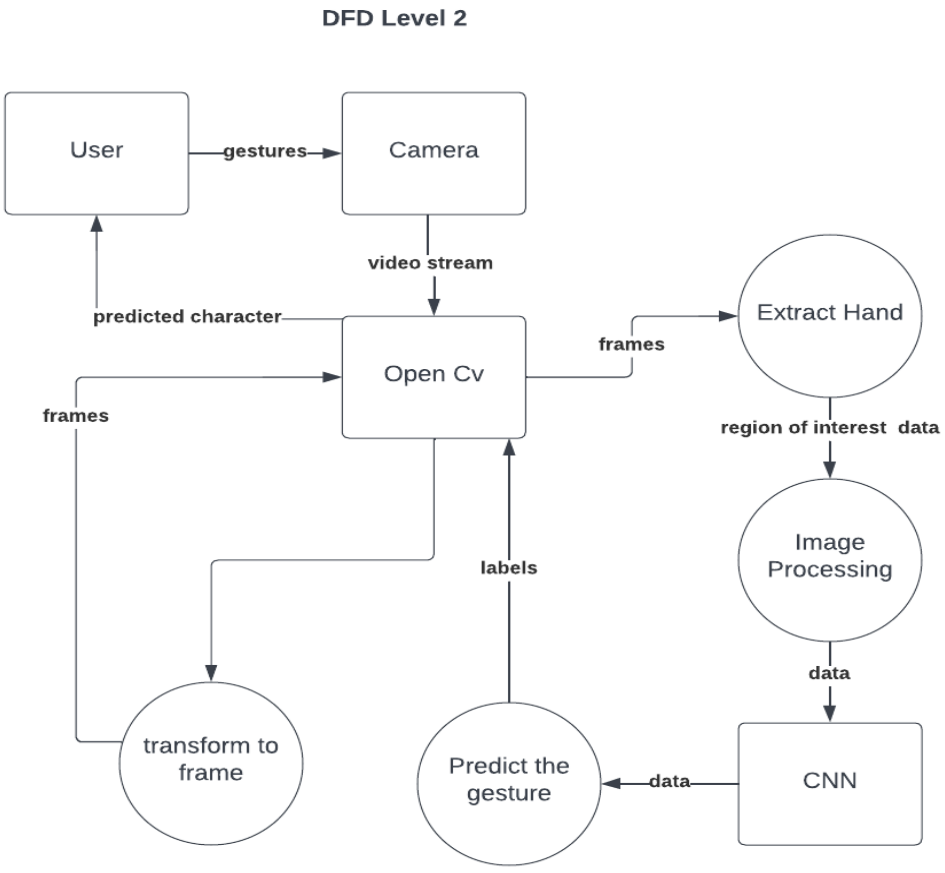
2. Scope of the Project:

This System will be Beneficial for Both Dumb/Deaf People and the People Who do not understands the Sign Language. They just need to do that with sign Language gestures and this system will identify what he/she is trying to say after identification it gives the output in the form of Text as well as Speech format.

4.0 Methodology:

Data Flow Diagram:

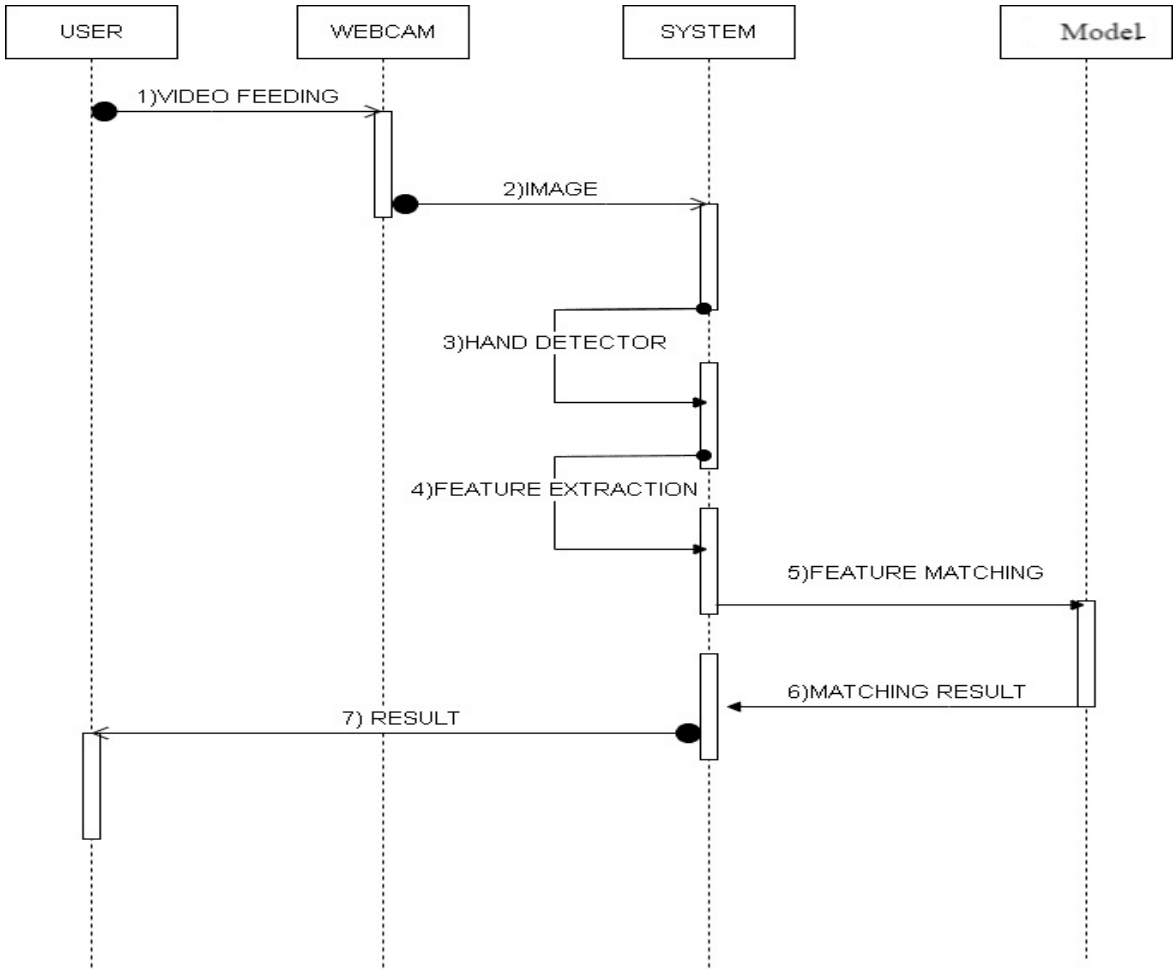




Use case Diagram



Sequence Diagram



5.0 Hardware and software to be used:

Sr. No.	Name of resource	Specifications
1.	Hardware	Computer System with 1. Storage of 2TB HDD and 256 GB SSD 2. RAM 8GB 3. Windows 10 pro
2.	Software Tool	PyCharm Environment
3.	Server	IIS Windows server
4.	Front End (GUI) Tool	Tkinter
5.	Back End Tool	OpenCV, Media-pip

6.0 Conclusion:

In this project, a functional real time vision based American sign language recognition for deaf and dumb people have been developed for ASL alphabets. We can achieve final accuracy of 98.74 percent on our data set. We would be able to improve our prediction after implementing two layers of algorithms in which we verify and predict symbols which are more similar to each other. This way we are able to detect almost all the symbols provided that they are shown properly and easily able to convert it to speech, there is no noise in the background and lighting is adequate